©2012-2018, GE Healthcare

5. Communication: The PH subsystem provides the functionality for the patient to communicate with the operator as well as the operator to communicate with the patient.
6. Emergency Shutoff: The PH system provides interfaces to stop scanning or remove power from the system in case of emergency.

4.10.1 Operator interface

The primary operator interface when physically preparing a patient for an exam is the patient table. There are pedal controls at the end of the table farthest from the magnet and controls on either side of the table closest to the magnet. The controls at all three locations allow the table height to be lowered to allow easy access and to be raised to reach the level required to be moved into the magnet bore. The raise/lower capability is provided via hydraulic pump when not docked to the system or via an automated electrical method when docked. At the end farthest from the magnet, functionality to mechanically and electrically dock the table to the magnet dock is provided.

Once the table has been raised to the proper height and docked to the magnet, the operator is provided with and identical set of controls on either side of the magnet next to the patient table that controls cradle position relative to the isocenter of the bore, laser light, setting landmark, bore lighting and bore air.

Most DV products provide an in-room display (IROC). The primary function of this monitor is to indicate the position of the cradle, the coil(s) that is connected and the signal from peripherals such as cardiac or breathing monitoring. A second function, coupled with a trackball/buttons located with the controls at either side of the table, provides the capability to enter patient weight and change the information being displayed (select from the various tabs). The MR450w product offers an option with a simpler display and no control.

Emergency stop and emergency off buttons are located at the operator console, on the magnet, and in other easily accessible locations defined by the site during system installation.

4.10.2 Patient interface

Elements of the patient handling design that have direct contact to the patient include bore ventilation and lighting, audio communications between operator and patient, a patient controlled alarm and support for physiological monitoring (ECG, respiratory).

The patient airflow is method to manage patient comfort. Air being blown down the patient bore and over the patient provides some cooling, provides a perception of cooling and reduces the claustrophobia effect. The operator has manual controls in the control panel on the magnet to adjust airflow. There are three different speeds as well as being able to turn the airflow off. For the 3T DV systems, the airflow is turned on at the highest speed when the predicted SAR is greater than 2.0 W/kg.

Similar to the airflow in the patient bore, the PH system provide control of bore lighting. Having the bore lit can also reduce the claustrophobia effect. The operator has manual controls in the control panel on the magnet to adjust the light level. There are three different light levels as well as being able to turn the bore light off.

The patient intercom provides two way audio communications between the patient and operator. The volume of the intercom can be set by the operator at the scanner console. Because gradient acoustic noise

©2012-2018, GE Healthcare

can overwhelm the patient's voice and render the intercom ineffective, the patient is provided a "squeezeball" to activate an alarm positioned near the operator console.

The in-room monitor is capable of displaying the physiological waveforms (e.g. ECG, pulsometer, respiratory) to assist in configuring these devices on the patient.

4.10.3 Patient transport

The patient transport design provides a removable rolling table, with vertical movement, and a patient positioning (or "landmark") system, which carries the patient into the magnet and aligns them at the scan position.

4.10.3.1 The patient table

The table is removable from the system to allow for remote patient prep and to assist in patient removal when necessary. This leads to several design considerations:

1. The table is designed to be easily maneuvered, even with a large patient on board. All four wheels are steerable, and the front (dock end) wheels may be locked to assist in control.
2. The table is designed to be stable when moving over uneven floors, to prevent tipping or patient falls.
3. The table is designed with vertical movement. It can lower to within 70cm of the floor, and raise to align for transport into the magnet.
4. The table is designed to allow the operator to release the table from the dock, fully manually. This allows the operator to quickly remove the patient under all conditions, including power failure.
5. All tables have a mechanical mechanism to dock to the system. For those tables that contain RF coil connectors or coils, there is also a mechanism to electrically connect to the system dock.

The maximum weight capability of the tables supported by the DV platform varies. In general, the tables associated with wide bore products can support more weight. The table below summarizes the specifications for the different products.

Product	GEM coil	static limit	Vertical limit
MR750w	Both (Y and N)	227 kg	227 kg
MR750	N/A	227 kg	160 kg
MR450w	Y	227 kg	227 kg
MR450w	N	227 kg	160 kg
MR450	N/A	227 kg	160 kg

Other features of the table are listed below.

1. The table allows for straps to be fastened along the table length. These straps can be used to position surface coils on the patient or as patient restraint.
2. The table provides "lift-able" safety side rails. They can be folded down to allow easy patient access, or raised to provide patient protection when needed.

3. The table provides an IV pole receptacle to safely carry a standard IV pole with the table.

4.10.3.2 Patient positioning system

The *patient positioning system* design provides precise, controlled, and automated movement for the patient in and out of the magnet bore (along the z-axis.) Major components are:

1. *The cradle*: supports and carries the patient into the bore. The cradle is designed to move smoothly along the z-axis while carrying the patient's weight. The cradle is designed to lock into position on the table, securing it while the table is undocked, and to lock onto the positioning system (described below) to allow movement along the z-axis. The cradle top is flat for the GEM table (contains the GEM PA coil or filler) and is curved for the non-GEM tables.
2. *The bridge*: supports the cradle as it moves along the z-axis. The bridge is designed to carry the patient weight, bridging between supports at the front and rear of the magnet, without allowing the bridge to contact the magnet enclosures.
3. *The cradle drive system*: moves the cradle to the desired position. The system consists of a rubber belt with teeth that is attached to a motor drive located in the rear pedestal. In the rear pedestal, there is a flag. When the position sensor hits the flag, this indicates that the cradle has reached the end of its travel. See below for more details.

4.10.3.2.1 Cradle drive system

The cradle drive provides motorized cradle movements, under either manual or automated control. During scan setup, the operator establishes a "landmark" for the scan by manually positioning the patient using a set of alignment lights. Upon operator request, the cradle drive moves this landmark position to the scanner's isocenter to prepare for the scan.

Features of the cradle drive system include:

1. Moving the patient from the landmark position to the scanner isocenter - The move distance is set by a system calibration (DQA), and the results are stored as the isoVectorZ parameter in the /w/config/MRconfig.cfg file. For a typical system, this distance is roughly a meter. The design is not intended to be accurate; a requested move of 1.000 meters does not assure the cradle moves exactly 1000mm, only that it moves the same distance every time.
2. Reposition the patient under system control – During an exam, the system can automatically adjust the location of the cradle such that the center of the imaging volume is at the isocenter of the system. This function allows the scanner to recenter the patient when the requested imaging volume is off-center along z or to move the patient for multi-position imaging so that the imaging volume is in the region that should produce the highest image quality (best shim, highest gradient linearity, least eddy currents, etc.).
3. Regulate the travel speed – In manual mode, the system provides two speeds (fast and slow). When the control button is slightly depressed, the cradle moves at the slow speed. When the button is depressed further, the cradle moves at the fast speed. The following parameters in the /w/config/MRconfig.cfg file correlate to the maximum cradle speed, acceleration and deceleration: tableMaxSpeed, tableMaxAccel and tableMaxDecel.

©2012-2018, GE Healthcare

4. "jog" feature: Allows the operator to move the cradle in 1mm increments to help with landmark positioning.

4.10.4 Magnet enclosure

The magnet enclosure provides for patient and operator protection and covers system components mounted on or near the magnet. Because of the magnet's size and prominence in the scan room, the design of the enclosures has a strong effect on the patient experience. The enclosure design must trade off tight fitting, sleek enclosures (desirable from aesthetics) against the need to mount system components within the enclosures (desirable for system performance).

The magnet enclosure is a large component of the external extent of system. Because of the proximity to the magnet, the enclosure is made from a non-magnetic material. In general, the endbells cover the space between the RF body coil and the magnet and extend about a foot radially. The top, sides and remaining from and back of the magnet are covered by the magnet enclosure. The enclosure consists of several distinct pieces that hang from brackets on the magnet. Separating into more than one piece allows the enclosure to be more manageable during installation and also minimizes the work required to access the components under the enclosure.

The DV platform has two enclosure types, one associated with DVApps (MR750, MR450 and MR450w without GEM) and one associated with MR450w with GEM and MR750w. The former is lightly colored while the latter is dark at the front of the magnet: black backing of the magnet controls with dark wood-grain finish and C-shaped lighting at the front of the magnet.

In MR750, MR450, and MR450w, magnet enclosure misalignment could affect the RF body coil positioning and alignment. This was tied to a variety of RF coil related problems. The MR750w design minimizes this effect by allowing for adjustable RF coil positioning independent of magnet enclosure alignment.

4.11 Cooling

The cooling subsystem provides thermal management for the system.

4.12 Patient safety

This section provides an overview of patient and operator safety concerns, and the mechanisms used by the system to control each. More detail is provided in ref [3], the *MR Safety Strategy Concept Document*.

The system controls the patient and operator exposure to five items:

1) SAR (RF power absorbed by the patient)
2) thermal environment seen by the patient
3) gradient dB/dt (exposure to changing gradient fields)
4) Static field characteristics (spatial dependence of the B0 field)
5) acoustic noise

Different types of controls are applied to different items. Some are controlled by design, some by software controls, and some have real time monitoring, as described below.

Practically all countries recognize IEC 60601-2-33 for regulating safety during MR exams. IEC 60601-2-33 recognizes several levels of patient exposure. The "normal" mode allows operation widely recognized as safe for the general population. "First control" mode allows higher exposure, and requires the operator to acknowledge such use. "Second control" mode is for research uses.

4.12.1 SAR control

Specific absorption rate, or SAR, is defined as the rate of RF energy dissipation in a patient, divided by the weight. This is a broad definition; more detail is needed, which is the purpose of this section.

MR scanners use RF power pulses to generate electric and magnetic RF fields. The patient's body is exposed to the pulses' magnetic and electric fields, which causes energy to be dissipated in the body. This energy is dissipated in the tissue exposed to the fields, which causes heating in the tissue. This heating should be limited, to keep the patient both safe and comfortable.

Even though the primary reason for concern is *tissue heating*, the regulatory agencies and manufacturers have chosen to focus on SAR as a *mechanism for controlling heating*. The rationale is that power dissipation is an input to the tissue heating process; since there a no known methods to effectively monitor the temperature of tissue during an MR scan, controlling and monitoring SAR is a means to the same end.

SAR is a rate of energy dissipation, divided by mass. This is energy per unit time, divided by mass, which simplifies to power/mass. Units of W/kg are commonly used.

SAR is a function of the tissue properties (conductivity and density) and the electric field. All of these vary across the body, especially the electric field, which is largely a function of the RF transmit coil. Therefore, SAR varies across the body, often by at least an order of magnitude.

Starting from the first MRI scanners, SAR has always been regulated (by the FDA, the IEC, or other regulatory agencies) and controlled (by both preventative and monitoring and controls built into GE scanners). The first

©2012-2018, GE Healthcare

scanners applied very conservative SAR limits – as lows as 0.4W/kg. Over time, higher levels of SAR have been permitted, based on accumulated clinical experience. Much, if not most, clinical scanning is done today at 4W/kg (averaged over the whole body, and over a six minute time – more on this below).

SAR increases with B0 field strength. For comparable systems (same FOV, similar RF transmit systems), SAR increases as the square of field strength. SAR is a concern at 1.5T, a major consideration at 3T, and imposes severe constraints at 7T.

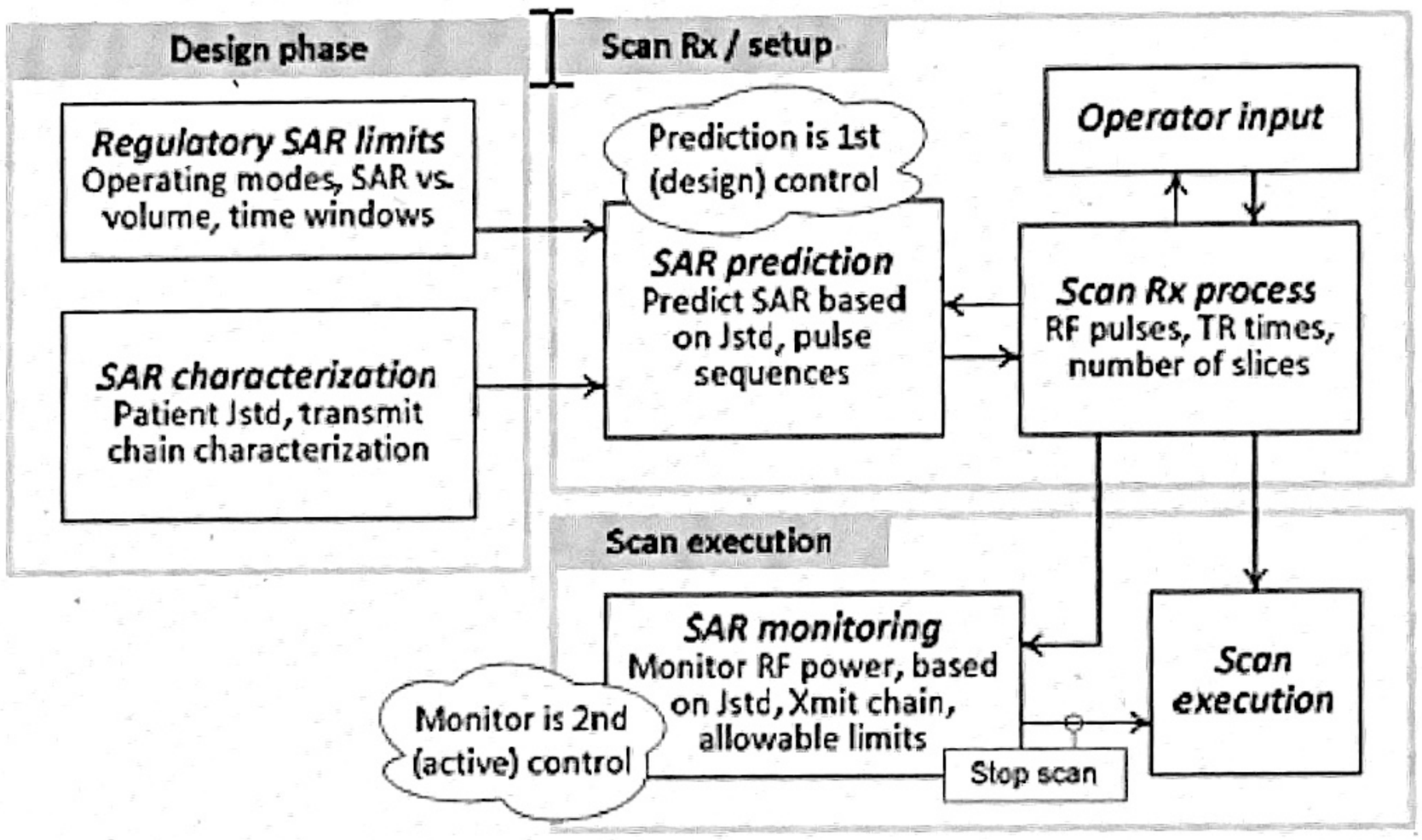


Fig. 4–32: SAR control design

The diagram above shows the design used for SAR control on the DV750w/750/450w/450 platforms. There are three main parts, outlined by the blue boxes:

1. Design phase work, including choosing regulatory limits, characterizing patient power absorption and transmit chain parameters, and simulation or modelling to understand parameters that are difficult to measure.
2. Scan Rx / setup processes. These are real time, interactive processes run during scan prescription. This is our primary SAR control, which predicts SAR based on the design-phase characterizations and the operator's input. Once the scan Rx's predicted SAR is within the SAR limits, the scan can proceed.
3. Scan execution monitoring. This RF power monitoring is our secondary SAR control, and will inhibit scanning if excess RF power is present.

The following sections give more details about how the individual functions are designed and operate:

Section	*Covers*
0	SAR Characterization
4.12.1.2	SAR Regulations

4.12.1.3 SAR Prediction (primary control)
4.12.1.4 SAR Monitoring (secondary control)

4.12.1.1 SAR characterization

For MR related purposes, SAR can be calculated as:

$$\mathrm{SAR} = \int_{\mathrm{sample}} \frac{\sigma(\mathbf{r})|\mathrm{E}(\mathbf{r})|^2}{\rho(\mathbf{r})}\mathrm{d}\mathbf{r} \qquad (4.14)$$

Where

r = the location of the tissue sample

σ = the conductivity of the tissue

ρ = the tissue density

E = the RMS electric field

The base form of this SAR equation is fairly intuitive (and easy to remember) since it is analogous to the familiar power equals voltage squared over resistance equation, and resistivity is the inverse of conductivity:

$$P = \frac{V_{rms}^2}{R} \quad \text{(here } R \text{ = resistance)} \qquad (4.15)$$

Since all of these parameters (conductivity, density, and electric field) will vary with location, SAR varies with location. Although most volume transmit RF coils are designed to produce a homogeneous ***B*** field within the imaging volume, the same is not true of the ***E*** field for a variety of reasons.

The integration produces an *average SAR*, averaged over the sample volume. SAR can be averaged over a wide range of volumes. If averaged over the whole body, we call it "whole-body SAR"; if averaged over a small sample (arbitrarily chosen as 10 grams of tissue), we call it "local SAR". (Note: The term "local SAR" is easily confused with "SAR for local transmit coils", which is a very different thing. The latter is described in §4.12.1.2.2.)

Although eqn. (4.14) provides insight into the mechanism for SAR, by itself it does not give a method for determining or characterizing SAR. In practice, SAR is characterized by two methods: measurement and modelling.

4.12.1.1.1 Average SAR Measurement

Average SAR is characterized by RF dissipation in the body, divided by the mass of the body or the portion of the body exposed to the RF fields. This process has several steps:

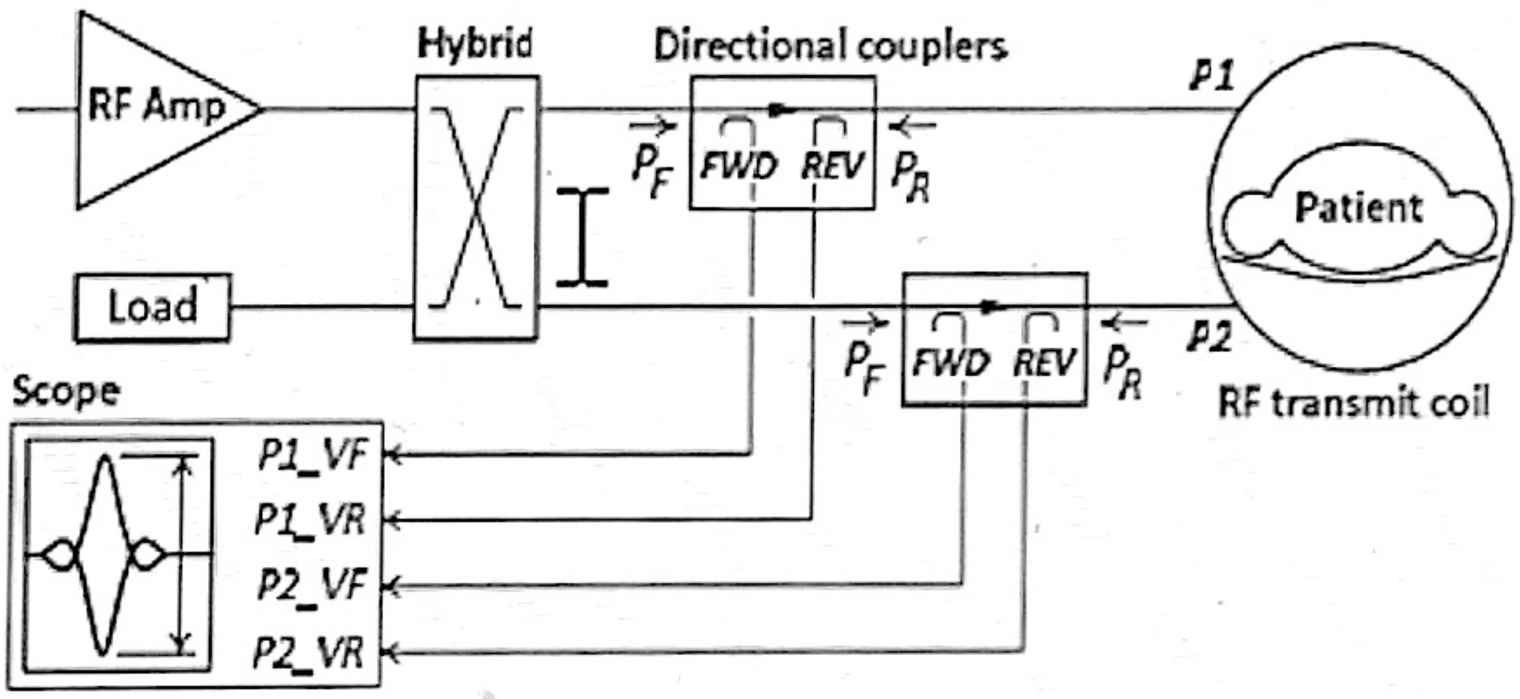
1. Measure the RF power dissipated in the transmit coil and body at a known B1.
2. Translate these RF powers to energy absorption, relative to a standard RF pulse.
3. Separate the energy absorbed by the transmit coil (Jcoil) and by the body (Jstd).
4. Fit the energy absorption of the body as a function of body mass, giving Jstd(w).

Note that the order of steps 2 and 3 can be reversed – they are basically independent.

Measure RF dissipation in the transmit coil and body

©2012-2018, GE Healthcare

The most common approach to measure RF power dissipation in the body uses directional couplers to sense the RF power flows into the RF transmit coil. A typical test setup is shown below, for a quadrature transmit volume coil in a hybrid based system. Other transmit architectures can use similar setups.



In this setup, directional couplers are used to sense the forward and reverse power flows into the two ports of the RF transmit coil; the total power absorbed by the coil is the sum of the four power flows. Some of this power is dissipated in the coil, and some in the patient. Power is measured on the scope at the peak of a RF pulse, *relative to a "known" B1 value.*

Since B1 varies across the body, and across phantom types, setting this "known" B1 value is difficult. For SAR characterization purposes, autoprescan is used to set the B1 value. Since APS is used in the vast majority of scans, this characterizes absorption with the same transmit B1 adjustment method most often used by the scanner.

Translate RF power to standard pulse energy absorption

Although a wide variety of RF pulse shapes are used in MR imaging, all of them can be characterized in terms of an arbitrarily chosen standard pulse. Our "standard" pulse is a 1ms wide rectangular pulse, with 11.74uT amplitude.

Why 11.74uT? For a rectangular pulse, 11.74uT gives a π radian (or 180 degree) flip angle.

$$\pi = \gamma \cdot B1 \cdot \Delta t \quad \rightarrow \quad B1 = \frac{\pi}{\gamma \cdot \Delta t} = \frac{\pi \text{ rad}}{2\pi \cdot 42.576E6 \frac{\text{rad/sec}}{\text{T}} 0.001\text{sec}} = 11.74\mu\text{T} \qquad (4.16)$$

Separate transmit coil absorption from body absorption

Fit the body absorption as a function of body mass

4.12.1.1.2 Local SAR Measurement

Local SAR is difficult to measure directly, but can be estimated by measuring the temperature rise in a sample or phantom of known heat capacity. (This approach exploits the close relationship between SAR and heating; remember, SAR was originally adopted as a proxy for heating.)

©2012-2018, GE Healthcare

Given a few assumptions (below), the rate of temperature rise in a sample can be found by balancing the energy dissipated in the sample with the energy stored due to increased temperature:

$$SAR \cdot m \cdot \Delta t = m \cdot c \cdot \Delta T \quad \rightarrow \quad \frac{\Delta T}{\Delta t} = \frac{SAR}{c} \tag{4.17}$$

Where

m = the mass of the sample (note that this drops out)
SAR = the SAR exposure of the sample
c = the specific heat of the sample
ΔT = the temperature rise
Δt = the duration of the exposure

This says that the rate of change in temperature is proportional to local SAR, and inversely proportional to the specific heat capacity. No other factors are involved. So for a uniform phantom, *dT/dt* is directly related to local SAR.

The biggest assumption is that there is no outflow of energy from the sample, or heat flow within the phantom – no conduction, no convection, etc. In practice, this requires that the sample be gelled to avoid liquid flow, and that the test be short enough that unwanted cooling or heat transfer does not take place.

The "short enough" constraint can be difficult to meet. If the phantom is water based, the specific heat is roughly 4.2 kJ/kg. At a dissipation of 10W/kg, it takes 420 seconds (7 minutes) to raise the temperature 1C. This small temperature rise would be hard to measure accurately, and the thermal time constant for the sample would have to be two or three times longer than seven minutes.

The point is you either need very high SAR levels, or good thermal isolation – or both – for this method to work well.

4.12.1.1.3 Modelling

Before around 2010, electromagnetic simulations of MRI transmit fields and their interactions with the human body were of limited capability, and not too widely used. After that, several factors converged, driving simulations to be both more capable and widely used. These factors were

a) the availability of the commercial EM modelling software,
b) the development of human body models with a range of sizes and both genders,
b) the computation power available to design engineers, and
c) the acceptance, and in fact expectation, by regulatory agencies that simulation be used.

By 2013, MR had added full body EM simulation to the new product development methods.

4.12.1.2 SAR regulations

Detailed SAR regulations are given in IEC 60601-2-33 Edition 3.1, section 201.12.4.103.2. These regulations are divided into two main parts: those applicable to *volume* transmit coils, and those applicable to *local* transmit coils. All transmit coils used in MR750w/750/450w/450 systems

©2012-2018, GE Healthcare

The IEC defines a "Volume RF Transmit Coil" as an "RF transmit coil ... that produces a homogeneous RF field over an extended volume encompassed by the coil." Examples include body, head, knee, all birdcage coils, and even a single loop coil, so long as it *encircles* the anatomy. "Local RF Transmit Coil" is any "RF transmit coil other than a volume transmit coil."

SAR is handled differently for volume and local transmit coils.

4.12.1.2.1 Volume transmit coils: whole body, partial body, and head SAR

For volume coils should provide a homogeneous transmit field. However, that field generally does not cover the whole body. This means that within a patient's body (or a phantom, for that matter) some regions will have higher SAR than others.

SAR is averaged over different volumes. The IEC defines three types of averaging:

1. *Whole body SAR* is defined as the SAR averaged over the entire body. For typical adults, the whole body is generally not exposed to SAR; the length of a body transmit coil is generally much less than the patient height. Still, the regulations treat SAR by averaging over the whole body.
2. *Partial body SAR* is defined over the volume of tissue exposed by a coil covering only part of the body. For example, an RF transmit coil for imaging the knee may only cover the knee plus some surrounding tissue; a 24cm FOV may be appropriate. The partial body SAR would be averaged over the volume of this coil's transmit field.
3. *Head SAR* is defined as SAR averaged over the head. In IEC regulations, the head is treated as a special and more sensitive partial volume case, with lower limits than would be permitted even for a whole-body average.

Note that the "whole body" averaging volume is clearly defined; it is the whole body. The "partial body" and "head" volumes are less well defined. This leaves interpretation up to the designers of the MR system.

The IEC table controlling volume transmit coil SAR limits is shown below – from Edition 3.1:

Table 201.105 – SAR limits for volume transmit coils

Averaging time	6 min		
	WHOLE BODY SAR	PARTIAL BODY SAR	HEAD SAR
Body region →	Whole body	Exposed body part	Head
Operating mode ↓	(W/kg)	(W/kg)	(W/kg)
NORMAL	2	2 – 10 [a]	3,2
FIRST LEVEL CONTROLLED	4	4 – 10 [a]	3,2
SECOND LEVEL CONTROLLED	>4	>(4 –10) [a]	>3,2
Long MR EXAMINATION specific absorbed energy	The maximum allowed specific absorbed energy is 14,4 kJ/kg (= 240 W min/kg) per MR EXAMINATION provided that the limits of this standard are still met		
Short duration SAR	The SAR limits over any 10 s period shall not exceed two times the stated values.		

[a] The limit scales dynamically with the ratio "exposed PATIENT mass / PATIENT mass":

NORMAL OPERATING MODE :

PARTIAL BODY SAR = 10 W/kg - (8 W/kg * exposed PATIENT mass / PATIENT mass)

FIRST LEVEL CONTROLLED OPERATING MODE :

PARTIAL BODY SAR = 10 W/kg - (6 W/kg * exposed PATIENT mass / PATIENT mass)

Fig. 4–33: IEC SAR limits for <u>volume</u> transmit coils

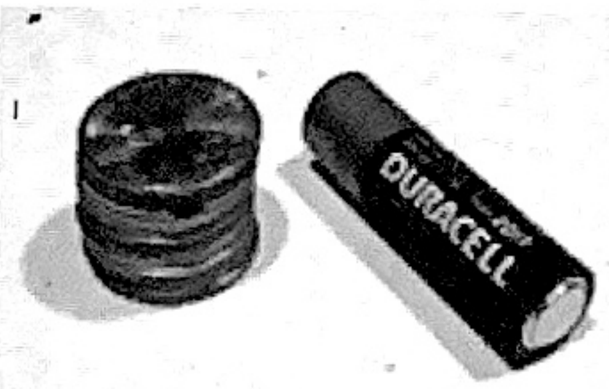
Example: An 80kg person that absorbs 200W of RF energy would have 2.5W/kg whole body SAR. The partial average SAR would be 5.0W/kg if the transmit coil covers only 40kg of the body.

4.12.1.2.2 Local transmit coils

Note! The terms "SAR for local transmit coils" and "local SAR" refer to very different things. "Local SAR" is described in section 0.

By definition, local transmit coils do not provide homogeneous transmit fields. Therefore, the SAR from a local coil would be expected to vary with location; some regions will have much higher SAR than others.

To deal with this variation, the IEC regulations limit local transmit coil SAR based on an average over a small sample of tissue, where the highest SAR occurs. "Small" is defined as a 10g sample of contiguous tissue, independent of sample's shape. Given typical tissue densities (like 1g/cc) this is a fairly small volume: the size of a stack of 12 US quarters or a bit larger than one "AA" battery.



Currently, the MR750/750w/450/450w products do not include any local transmit coils, nor do they include provisions for local transmit coil SAR control.

The IEC table for local transmit coil SAR limits is below – from Edition 3.1:

Averaging time	6 min		
	LOCAL SAR		
Body region →	Head	Trunk	Extremities
Operating mode ↓	(W/kg)	(W/kg)	(W/kg)
NORMAL	10 [a]	10	20
FIRST LEVEL CONTROLLED	20 [a]	20	40
SECOND LEVEL CONTROLLED	>20 [a]	>20	>40
Short duration SAR	The SAR limits over any 10 s period shall not exceed two times the stated values		
[a] NOTE In cases where the orbit is in the field of a small LOCAL RF TRANSMIT COIL, care should be taken to ensure that the temperature rise is limited to 1 °C.			

Fig. 4–34: IEC SAR limits for local transmit coils

4.12.1.2.3 Time averaging

Since MRI uses RF pulses, arranged into a wide variety of sequences, SAR varies over time as well as location.

Two time windows are commonly used for averaging SAR exposure: a 10 second short-term exposure window, and a long-term 6 minute window. An additional "long exam" limit is actually an *integrated* SAR limit, as opposed to a time averaged limit.

The six-minute window averaging window is the most familiar limit. Typically, when a scan is referred to as a "4W/kg scan", the reference is to the six minute time average, applied with a six minute wide rectangular or "boxcar" window.

The IEC limits "short duration SAR" (10 sec average) to no more than twice the applicable 6 minute averaging limits. If the 6 minute limit is 3.2W/kg, the 10 second limits is 6.4W/kg.

The "Long MR Examination" limit applies to the SAR integrated over the length of the examination. The limit is 14.4 kJ/kg. This would be a one hour scan at 4W/kg (3600 seconds x 4 W/kg = 14,400 J/kg) or a two hour scan at 2W/kg.

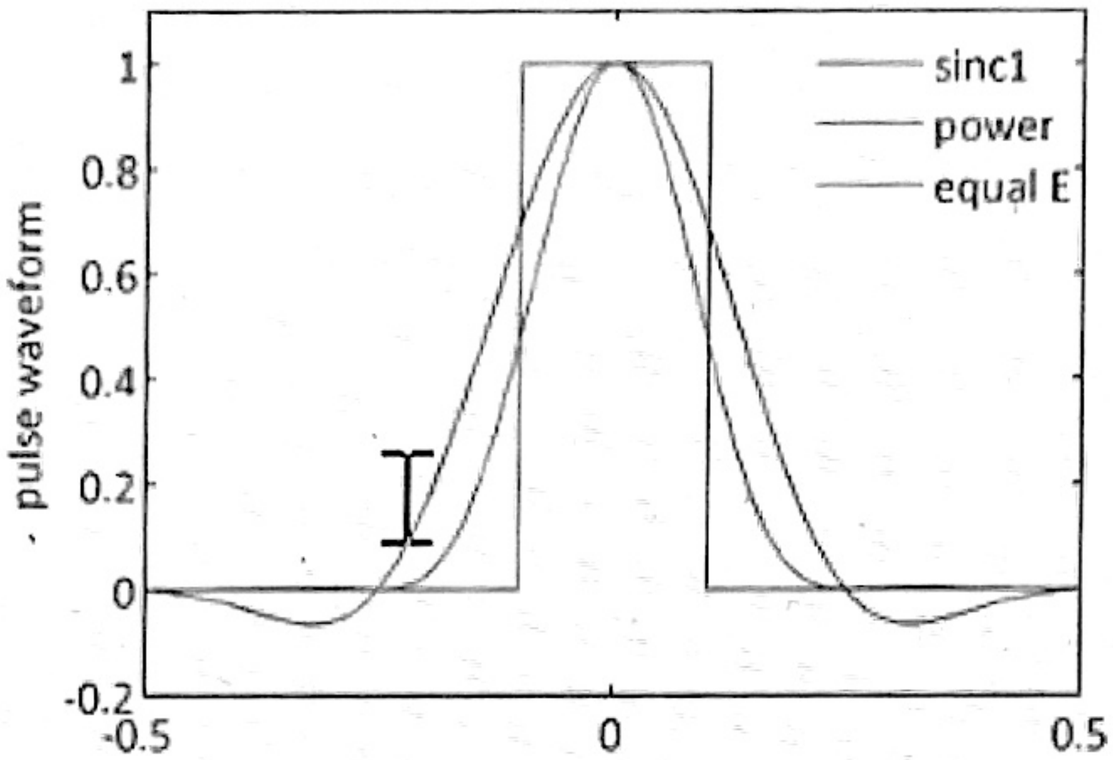
Note: We do not place controls on "instantaneous" SAR, which could be interpreted as the maximum SAR during an RF pulse. For example, a 100kg patient may absorb 10KW at the center of a 180 degree pulse, which would be 100W/kg averaged over the whole body. This high SAR spike must be surrounded by enough low SAR intervals to average down to the short duration SAR limit, when averaged over 10 seconds.

4.12.1.3 SAR prediction

As of 2014, "SAR prediction" is evolving rapidly. This section outlines the methods used in the MR750w/750/450w/450 scanners.

If *Jstd* – the energy absorbed for a "standard" 11.4uT 1ms rectangular pulse – is known, the energy absorbed for any sequence arbitrary pulses can be predicted.

Released



For a particular transmit configuration (transmit coil, patient size and position, etc.), a coefficient *K* can be defined to relate the energy absorbed to the integral of the square of *B1*:

$$E = K \cdot \int_{pulse} B1^2(t)dt \tag{4.18}$$

If this equation is applied to the standard pulse, the energy is *Jstd*. The pulse length *PWstd* and pulse amplitude is *B1std*. This allows *K* to be determined based only on the standard pulse characteristics:

$$K = \frac{Jstd}{B1std^2 \cdot PWstd} \tag{4.19}$$

Substituting this back in for K in the first equation gives and expression for the energy in any arbitrary pulse, based only on the characteristics of the standard pulse.

$$E = \frac{Jstd}{B1std^2 \cdot PWstd} \cdot \int_0^{PW} B1^2(t)dt = Jstd \cdot \frac{PW}{PWstd} \cdot \left(\frac{B1}{B1std}\right)^2 \cdot \int_0^1 \rho^2(t)\,dt \tag{4.20}$$

where

$\rho(t)$ = the pulse envelope, with a peak value of 1, and covering a time interval of 0 to 1.

This allows easy evaluation of the energy deposition for any pulse sequence. For each pulse envelope, the integral can be calculated once and stored. This stored value is scaled by the pulse's B1 and pulse width, giving the energy in that pulse. The total energy is just the sum of the individual pulses.

4.12.1.4 SAR monitoring

The system controls SAR by monitoring RF power amplifier output. By knowing the patient body mass (provided by the operator) and the percentage of RF amplifier output that is absorbed by the patient (determined by experimental characterization during the system design process), the system calculates SAR. This is compared to the applicable limits, and the scan process is inhibited if SAR exceeding the limit is observed.

©2012-2018, GE Healthcare

Local SAR control is based on computer models of the human body. During the system design process, computer simulations combine models of the transmit RF coil with human body models to determine the spatial distribution of energy absorption. By characterizing the ratio between the local SAR and whole body SAR, local SAR can be controlled based on whole body SAR control. The computer models must cover a range of body sizes, shapes, and positions to understand how local SAR distributions vary across the human population.

4.12.2 Thermal environment

The thermal environment surrounding the patient affects the patient's ability to dissipate heat and manage body temperature. As the patient thermal environment warms, it will be harder for the patient transfer heat to their environment, which increases the thermal stress on the patient. Therefore, controls are put on the thermal environment.

In some cases, regulatory agencies put limits on the thermal environment, either as directly (as limits on environmental temperature) or indirectly (by decreasing the allowable SAR limits as environmental temperature increases.)

The system uses environmental temperature (TE) as the basis for patient thermal environment controls. TE is defined as the average of bore temperature (TB) and ambient air temperature (TA). Ambient air temperature is controlled by the site operator, but monitored by the system via a temperature sensor. Bore temperature is controlled by limiting the gradient coil's power dissipation.

4.12.2.1 Humidity considerations

Relative humidity (RH), as well as temperature, affects the body's ability to shed heat. It is more difficult to stay cool in a high RH environment than at low RH.

Regulatory agencies include RH as a consideration in determining allowable SAR. As humidity increases, allowable SAR decreases.

The system does not have a humidity monitoring mechanism, so 100% RH is assumed in the system's SAR controls.

4.12.2.2 Gradient coil temperature controls

The system maintains two levels of control on gradient coil temperature, because GC temp plays a significant role in the patient's thermal environment.

The first GC temp control is a predictive control, implemented via limits on the power input to the gradient coil.

The second GC temp control is a monitoring control, implemented thru temperature sensors built into the gradient coil.

4.12.3 Gradient dB/dt

The changing gradient field can cause undesirable effects on the patient. These are usually called dB/dt effects or PNS effects (for peripheral nerve stimulation), and range from mild to severe. As dB/dt is increased, a tingling sensation becomes perceptible, followed by involuntary muscle twitching. At roughly 2X the tingling perception threshold, sensations of electric shock occur. (At roughly 10X the perception

©2012-2018, GE Healthcare

threshold, dB/dt stimulation can interfere with normal cardiac functions. The gradient power needed to reach this level is in excess of what the system's gradients can produce.)

Regulatory agencies limit dB/dt exposure to levels corresponding to mild effects. For normal mode scanning, roughly 5 to 10% or patients will experience some sensations. In first control mode scanning, this increases to roughly 50%.

The dB/dt regulations were solidified in the mid-1990s, when gradient subsystems became powerful enough to cause dB/dt stimulation.

There are three system design parameters that control dB/dt exposure: the physical size of the gradient field (larger gradient fields increase dB/dt exposure), the rate of change in the gradient (increased slew rates increase exposure) and the time duration of the gradient ramp (longer ramp durations increase exposure). In addition, the patient's body orientation and position have a strong effect of patient susceptibility to dB/dt exposure.

The system control dB/dt exposure by limiting the allowable gradient waveforms, taking into consideration the gradient coil design. This is implemented thru correct coding of the pulse sequences. There is no dB/dt monitoring during the scan.

4.12.4 Static field characteristics

The magnet's B0 field strength varies greatly at different positions in and near the magnet. Near to the magnet, the field can be strong, with a steep spatial gradient. This results in three separate safety concerns: 1) high fields, 2) steep field gradients, and 3) the combination of high field and steep spatial gradient, (commonly called the *force product*).

Regulatory agencies generally require characterization of all three qualities (often jointly called static field characteristics), for all user accessible areas.

There are two product design aspects that control the static field characteristics. First, the magnet design controls the B0 strength, and how fast the B0 field rolls off outside the bore. (In order to minimize the B0 field footprint, designers seek to have the field decrease rapidly, which can result in high field gradients.) Second, the magnet enclosures determine the user accessible areas. Since the B0 field rolls of quickly, the enclosures can prevent user access to high force product regions.

The characterizations are currently included in the system Operator Manual and the Safety Manual. In the future, when the safety effects of static fields are more clearly understood, they may be only included in the Operator Manual.

4.12.5 Acoustics

MR systems produce acoustic noise from a range of sources. By far, the most significant source is gradient noise. The primary mechanism is the gradient currents, combined with the B0 field; this produces forces on the gradient coil that lead to acoustic noise. A secondary mechanism is eddy currents on other conductors, which similarly lead to acoustic noise; RF body coils are particularly susceptible to this effect, due to proximity between the coil and the patient.

The IEC regulations limit both the peak level (Lpeak) and an average level (Leq).

©2012-2018, GE Healthcare

4.12.5.1 Lpeak

Lpeak is "the peak impulse sound pressure level (SPL) measured during the measurement period." It is a peak measurement; a single pulse would (theoretically) have the same Lpeak as a series of similar pulses.

The system is designed so that all scans will have Lpeak <140dB (relative to 20uPa.)

This is a very high SPL, roughly the irreversible damage threshold. (Most sources list the pain threshold at 130dB.) The Lpeak control is by design; given the stiffness and mounting of the gradient coil, and the power available from the gradient drivers, the system is not capable of exceeding 140dB sound pressure levels.

4.12.5.2 Leq

Leq is "the A-weighted rms SPL averaged over the measurement period". Leq is an averaged or integrating method, so multiple pulses result in higher Leq compared to a single similar pulse.

The system has controls to keep patient exposure Leq<99dBA, as outlined below. The rationale for 99dBA is described in the MR Safety Strategy Concept Document.

The controls consist of two parts:

1. The first control is recommended use of earplugs. The system Operator Manual recommends using earplugs for all patients; a variety of inexpensive and comfortable earplugs are readily available. Noise reduction ratings (NRR) of 30 are common. These would reduce 129 dBC (note that is dBC, not dBA) scanner noise to 99dBA at the patient's ear. If scanner noise is measured with A-weighting, the NRR must be reduced by 7, so NRR 30 plugs will offer only 23dB attenuation. To keep patient ear noise at 99dBA, the scanner noise must be less than 122dBA.
2. The second control is a design control on the allowable sequences. During system design, acoustic measurements, on a range of pulse sequences, characterize the system acoustic performance. Sequences that exceed the allowable scanner noise may be modified to lower acoustic noise.

See OSHA Noise and Hearing Conservation Program for a reference.

5 Performance

Expected content:

This section defines the performance expected from the design. This should include what parameters are tested, the high-level test methods, and the expected performance. The specific procedures for performance testing should be in a separate test plan document.

*Note that design documentation is used for **engineering design test**. Therefore, it may be appropriate to have tighter performance expectations than those enforced by the requirements. You may also want to call for tests across multiple samples, under non-standard conditions, etc.*

5.1 Overview

Expected content: Performance of the overall system described in the document.

©2012-2018, GE Healthcare

Instructions: If some performance specs are described in a related requirements document, they can be repeated here for reference. They should be clearly marked as for reference only, with a note giving precedence to the requirements document in case of conflict.

5.2 System performance

5.2.1 SNR

5.2.2 Ghosting

5.2.3 Uniformity

5.2.4 MR signal stability

5.2.5 Eddy currents

5.2.6 Gradient strength and slew rate

5.3 Magnet performance

5.4 Gradient performance

5.5 Transmit chain performance

5.6 Receive chain performance

5.7 RF coils performance

6 Revision history and document information

Formal revision control is implemented by storage in MyWorkshop DOC0700133.

The table below lists changes made since the most recent formal release, with a short description of the when, who, what, and why of the change. Place most recent changes at the top of the list.

Date	*Change by*	*Description*
2017-OCT-10	MGE	Added several sections under 4.6.3, *Gradient Coils*, covering Rio HRMw GC.
2017-MAY-31	MGE	Added 4.6.1.4, information on gradshim and gradshim resolution. (Added since Rio is adopting 20-bit gradshim resolution.)
2017-MAY-31	MGE	Added 4.4.10.1, on Rio's gradient switching induced body channel coherent noise causes.
2017-MAY-24	MGE	Added information on spherical harmonics; general info in Appendix B3, and how they are used in gradwarp sections.
2017-MAR-31	MGE	Added 4.4.10, on coherent noise.
2012-APR-01	MGE	Extensive updates

©2012-2018, GE Healthcare

Appendix A Variable Names and Glossary

A1 Variable Names

A list of common variable names used in the document. This is not a complete list; some limited use variables will be described locally.

Conventions

Many variable names may either apply to all axes or to a specific axis. A subscripted *x*, *y*, or z indicates application to a specific axis. Example: a 10 cm *FOV* implies a field of view that is 10 cm in all directions. A 10 cm FOV_X implies a 10 cm field of view in the *x*-direction only.

Values for variables that apply to more than one axis may often be given as ordered pairs or triplets. Example: G = (3.2,3.2,4.0) G/cm is shorthand for G_X = 3.2 G/cm, G_Y = 3.2 G/cm, G_Z = 4.0 G/cm.

Name	*Description*
B0	The main static magnetic field. A 1.5 T system has a 15000 Gauss *B0* field.
B1	The RF transmit magnetic field. A 1.5 T system has a 63.86 MHz *B1* field.
$B1_{RMS}$	The root-mean-square value of the *B1* field. Inherent in any *RMS* specification is an averaging period, so this needs to be clear in the context.
FOV, FOV_X ...	Image FOV. A 12 cm *FOV* image covers 144 cm^2, over a 12 cm by 12 cm square.
FOV_S	System FOV. This is the specification volume for the system.
G, G_X ...	Gradient field, or gradient field strength
GFS, GFS_X ...	Full scale gradient field strength.
I_G, I_{Gx} ...	Gradient current
I_{GFS}, I_{GFSx} ...	Full scale gradient current
J_{COIL}	The amount of RF energy dissipated in an RF transmit coil while creating a standard RF pulse.
J_{STD}	Joules per standard pulse. The amount of RF energy absorbed by a human subject during a standard RF pulse; it is a function of the subject's weight and body shape, but usually expressed only as a function of weight.
J_{STDM}	*Mean* joules per standard pulse; the joules per standard for the *average* human subject.
K_{GC}	Gradient coil gain, in gradient coil field produced per Amp of gradient current.
SR, SR_X ...	Gradient field slew rates.
T1	Spin-lattice relaxation time
T2	Spin-spin relaxation time
TE	Echo time
TR	Sequence repetition time
T_{GFS}, T_{GFSx} ...	Gradient transition time, from zero to full scale gradient.

©2012-2018, GE Healthcare

A2 Glossary

The acronyms and terms listed below are in addition to these described in the *MR System Glossary*. Because some terms may have multiple Glossary entries, the reader may need to determine the correct usage from the context.

Term	*Description*
APS	Automatic Prescan
BRM	Body Resonator Module: *a gradient coil, c. 1996, with full body FOV*
CRM	Cardiac Resonator Module: *a gradient coil, c. 1998 (?), with smaller FOV*
DFT	Discrete Fourier Transform
EC	Eddy Current
ECC	Eddy Current Compensation
FFT	Fast Fourier Transform: *mathematically equal to the DFT, but much more efficient to compute.*
GC	Gradient Coil: *the GC converts the output current of the GD into a gradient field.*
GD	Gradient Driver: *the GD converts the output command waveforms from the SSC into gradient current waveforms.*
GFS	Gradient field strength at Full Scale
GEHC	GE Healthcare
HNU or HNS	Head/Neck/(Spine) Unit. A multiple receive coil array for head, neck, and upper spine imaging.
MPS	Manual Prescan
MR	Magnetic Resonance
MRI	Magnetic Resonance Imaging
NIST	National Institute of Standards and Technology
NMR	Nuclear Magnetic Resonance
PA	Posterior Array. A multiple receive coil array for spine and torso imaging.
PS	Prescan
RF	Radio Frequency
RFPA	RF Power Amplifier
RFPM	RF Power Monitor
RFTL	RF Transmit Load
R1, R2	Receive gain 1 (analog gain, applied before receive signal digitization) and receive gain 2 (digital gain, applied after digitization).
RM	Resonator Module
ROI	Region Of Interest: *an area inside an image, typically selected for analysis*
RXC	RF Receive Chain
SAR	Specific Absorption Ratio: the energy absorbed by a patient or part of the patient when exposed to RF.
SR	Gradient field Slew Rate

©2012-2018, GE Healthcare

SSL	System Safety Limit
SNR	Signal to Noise Ratio
SSC	System Sequence Control
Standard RF pulse	A rectangular RF pulse, 1.00ms long, 11.74uT amplitude. The "standard" pulse has a 180 degree flip angle and the same energy as a 3.2ms sinc1 pulse.
TG	Transmit Gain
TRM	The Twin Resonator Module: a dual FOV gradient coil, built with a large FOV coil (WHOLE) and a small FOV coil (ZOOM)
TXC	RF Transmit Chain
UPM	The Universal Power Monitor is an implementation of the RFPM function, c. 2003 (?).
UNBLK	The Unblank signal controls the system's T/R mode.
XRMB	eXtreme Resonator Module BRM: a gradient coil design, c. 2007, capable of full body FOV.
XRMW	A wide bore (70cm) gradient coil, a successor of the XRMB.

Appendix B Background information

This appendix contains information useful to MRI system design, but not specific to it. These topics are useful to a wider range of applications or industries, so there is a wealth of information available; the content below tends to filter that to the information most useful to MRI.

B1 SNR

A general and fundamental definition for signal to noise ratio, or *SNR*, is used over a wide range of subjects: it is the ratio of a dataset's mean, μ, to the standard deviation, σ:

$$SNR = \frac{\mu}{\sigma} \qquad (C.1)$$

For images, the "dataset" is the pixel values inside a region of interest, or ROI:

$$SNR = \frac{S}{N} = \frac{\mu_{ROI}}{\sigma_{ROI}} \qquad (C.2)$$

B1.1 Measurement methods

The "mean over stdev" definition can be applied directly, and will give reasonable results assuming that *all of the variation in the pixel values is due to random variations*. For most images, unfortunately, this assumption is *not* true; there may be image shading, phantom structure, or other non-random variations.

A number of industry standard image SNR measurement algorithms are used to deal with these non-random variations.

Below is a summary of these methods. Note that all methods share a common method for calculating *S*, the signal; it is the mean value of the pixels in the ROI. The calculation of noise *N* is what changes between methods.

Note: NEMA Standard MS-1 describes image SNR measurement procedures in detail. The four general methods outlined below match the four NEMA methods of MS-1. In addition, NEMA document defines standards for ROI size, shape, location, etc. However, the general methods can be applied to broader range of problems than the specific NEMA tests.

B1.2 Subtracted images method

Note: The image subtraction tool provided on GEHC MR scanners may rescale the subtracted image, without notifying the operator. This leads to incorrect *SNR* results; therefore, it should not be used to subtract images for *SNR* testing.

This method calculates noise from a "subtracted image", created by taking the difference between two images. The difference image should have a mean of zero, and a standard deviation higher by $\sqrt{2}$ compared to the base images. Therefore, the base image noise is calculated from the subtracted image noise:

$$N = \frac{\sigma_{subtracted}}{\sqrt{2}} \qquad (C.3)$$

©2012-2018, GE Healthcare

This method considers *any* non-repeatable content in the image as part of the noise. Therefore, it is useful for characterizing overall system performance; gain stability, random ghosting, and other signal variations will be included in the "noise" measurement. Other methods may be preferable for if this is not the intent.

This method is the basis of NEMA method 1. The NEMA documentation describes methods to remove effects of artifacts, especially low spatial frequency artifacts like shading. These methods should not be used without a full understanding of how they affect the measurement.

B1.3 Noise-only image method

This method calculates noise from a noise-only image. The standard practice is to disable RF transmit and rescan, so no MR signal is produced; the resulting image has the same scan conditions, but contains only noise.

By default, the system displays magnitude images. In a noise-only magnitude image, the pixels in the noise image should follow a Rayleigh distribution, and the base image noise can be calculated from the standard deviation of the noise-only image:

$$N = \frac{\sigma_{noise\ image}}{\sqrt{(4-\pi)/2}} \approx \frac{\sigma_{noise\ image}}{0.655} \qquad (C.4)$$

The 0.655 factor results from the Rayleigh distribution statistics.

This method is the basis for NEMA method 2.

B1.4 Special image method

This method uses a specialized acquisition technique to minimize the effects of scanner instabilities or other temporal changes. A raw data set is acquired that can be split into two halves and reconstructed into two images, which are then subtracted.

This method should not be used without a clear rationale of why it was chosen over the alternatives. It forms the basis for NEMA method 3, and is included here for completeness.

B1.5 Background noise method

This method calculates image noise from an ROI place in the image background. Care needs to be taken to avoid low level artifacts (like smearing or ghosting) that would contaminate the background with signal artifact. Assuming a magnitude image is displayed, the background noise will follow a Rayleigh distribution, as in method 2, so the noise calculation is identical:

$$N = \frac{\sigma_{noise\ image}}{\sqrt{(4-\pi)/2}} \approx \frac{\sigma_{noise\ image}}{0.655} \qquad (C.5)$$

A quick check for signal contamination in the background noise is to compare the mean and standard deviation for the noise only ROI. Since the pixel intensities should be Rayleigh distributed, the ratio between mean and stdev should be close to the theoretical value of

$$\frac{\mu}{\sigma} = \sqrt{\frac{\pi}{2}} \approx 1.253 \qquad (C.6)$$

sed

©2012-2018, GE Healthcare

If the ratio is outside of the 1.15-1.35 range, the background noise probably is not Rayleigh distributed; an alternative method should be used to confirm the method 4 results.

NEMA method 4 uses this method.

B2 Noise Figure

Noise factor F measures the degradation in signal to noise ratio caused by the system receive chain, compared to an ideal receiver; noise figure NF is simply noise factor expressed in dB.

$$F = \frac{SNR_{in}}{SNR_{out}} \quad \text{and} \quad NF = 10\log F \tag{D.1}$$

SNR_{in} is the signal to noise available at the system's input (generally taken as the preamplifier input) and SNR_{out} is at the system output (generally taken as an image). A system with F=1 (a 0dB noise figure) would be a perfect system; there would be no degradation in signal to noise anywhere in the system. A system with F=2 (a 3dB noise figure) has lost half the available SNR, due to imperfections in the system.

Noise factor and figure are relatively recent concepts; they date from the 1940s when Harold Friis created the definition above. Historically, "noise figure" was used for both linear and dB expressions. Modern usage is "noise factor" for the linear expression, and "noise figure" for the dB version, to avoid the obvious confusion.

If the system gain G is known, the equation above reduces to

$$F = \frac{SNR_{in}}{SNR_{out}} = \frac{S_i/N_i}{S_o/N_o} = \frac{S_i/N_i}{S_iG/N_o} = \frac{N_o}{GN_i} \tag{D.2}$$

Note that the output noise N_o is *not* equal to the "gained up" input noise N_iG, as the system always adds additional noise.

This relationship is generally impractical to measure, as it requires very accurate measurements of gain G and input noise N_i. But it does make clear that noise figure is just the ratio between the system's actual output noise N_o and what would be expected based on input noise N_i and system gain G.

Measuring F based on these definitions is not very practical, as determining SNR_{in} is hard; the signals are tiny, and there is no practical way to quantify noise at this level. Several practical alternatives exist.

B2.1 The *Y*-factor method

One practical alternative is called the Y-factor method. A noise source, capable of switching between two calibrated noise levels (typically called "on" and "off", and related by an *excess noise ratio*, or *ENR*), is fed into the system input. The output noise of the system is measured with the noise source at both levels. From the output noise measurements and *ENR*, the noise factor can be found.

The *Y*-factor is defined as the ratio of noise *power* at the system output, for the noise source on (N_2) and off (N_1). For an image, the noise power is the variance in pixels of a noise image. To get the variance, square the standard deviation of an ROI obtained from the noise on and noise off images.

$$Y = \frac{N_2}{N_1} = \left(\frac{\sigma_2}{\sigma_1}\right)^2 \tag{D.3}$$

The noise factor of the system is based on Y and *ENR* of the noise source: